



Introduction to HPC and Parallel Programming

Karuna Prasad
Scientist E
C-DAC, Bangalore
karunap@cdac.in

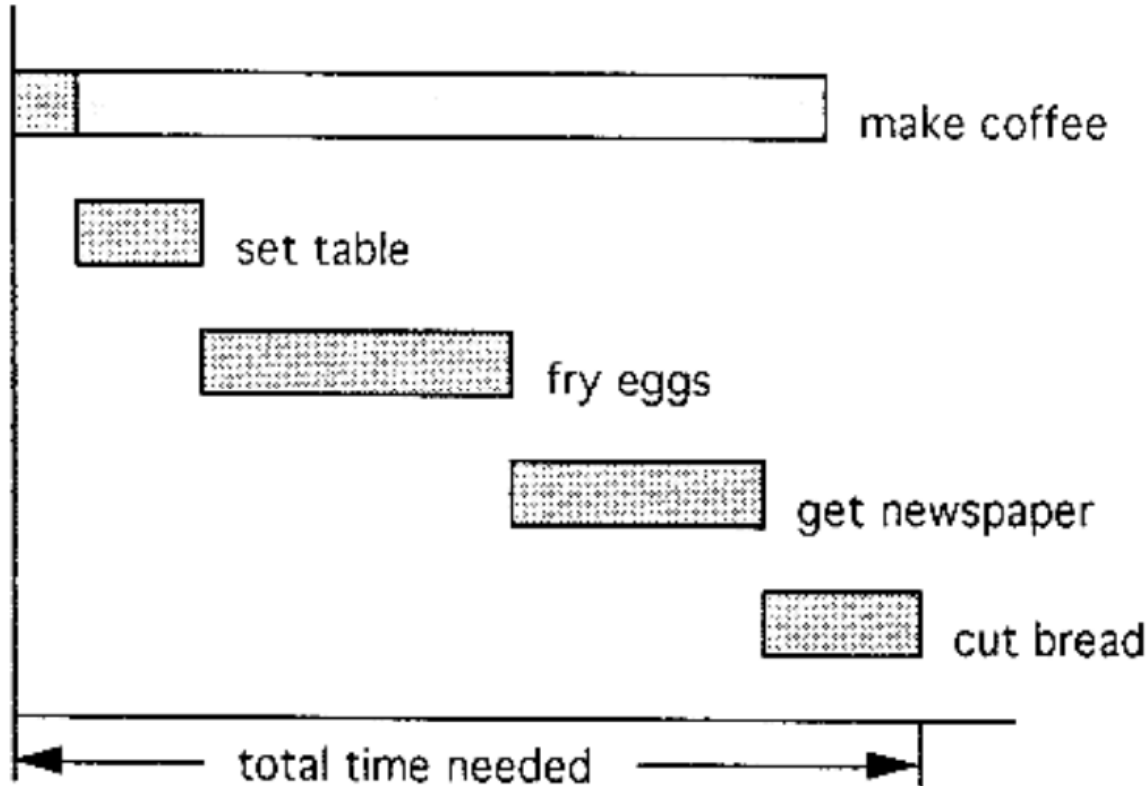


Agenda

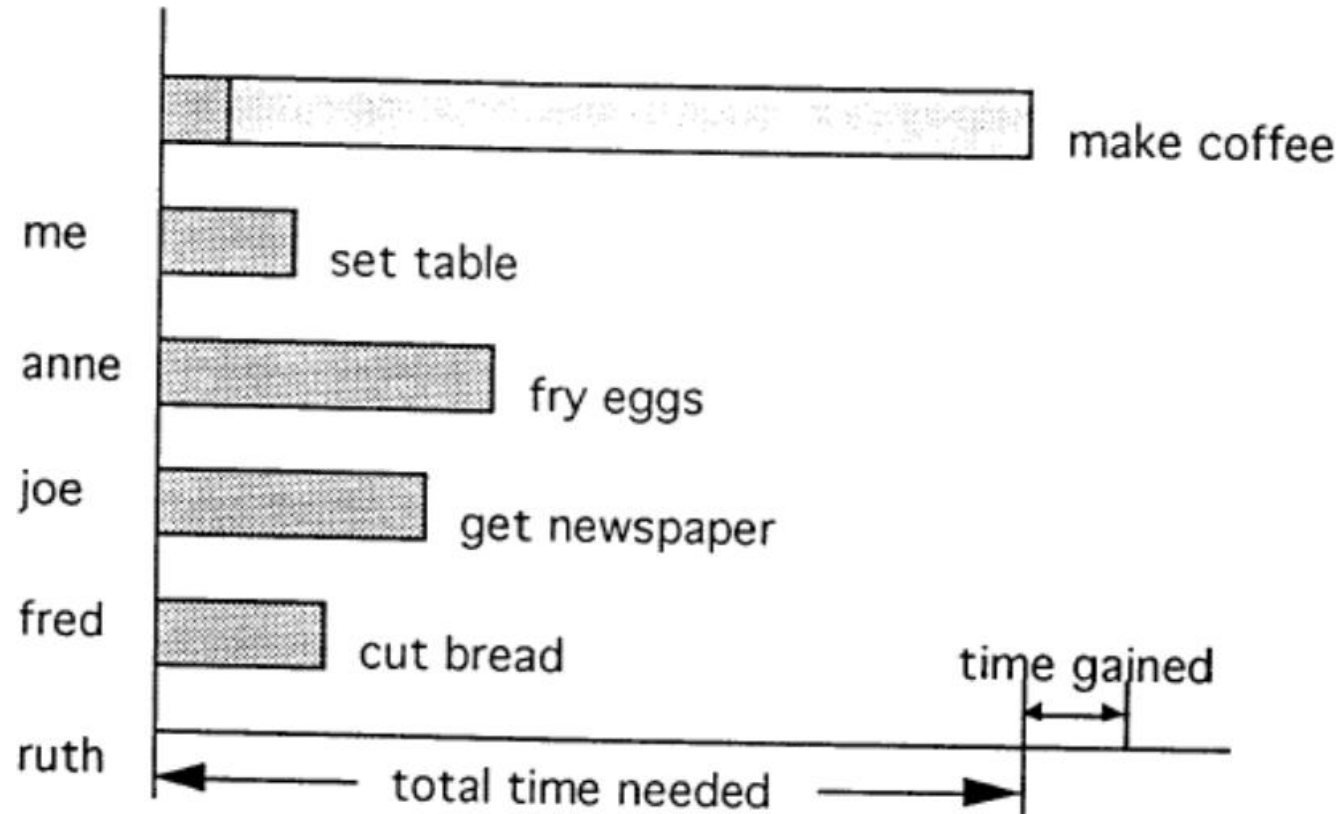


- Parallel Computing
- Parallel Architectures
- Memory Architectures
- Flynn's Taxonomy
- High Performance Computing in India
- NSM- Param Utkarsh Supercomputer
- Parallel Programming Models

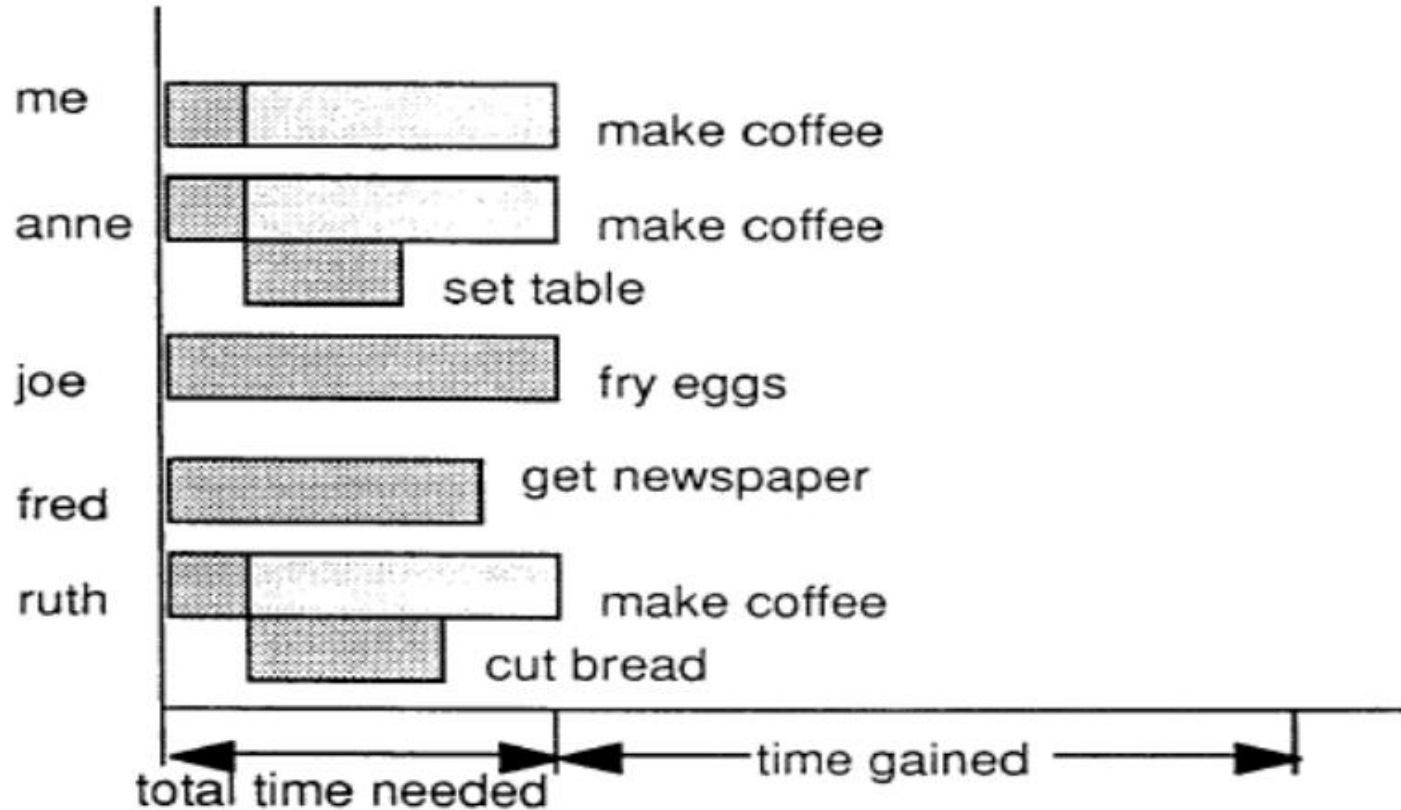
How I used to make breakfast....



How to set family to work



How finally got to the office in time....





Serial Computing



- If 1 person takes 15 hours to complete a job
- How long will it take if 15 person work together



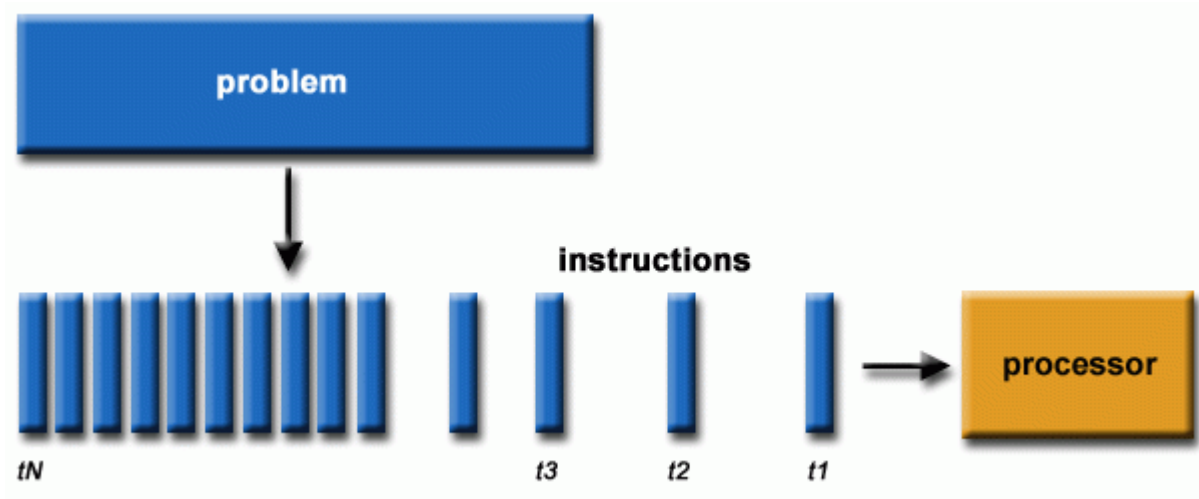
Parallel Computing

- Save time - wall clock time
- Solve larger problems
- Provide concurrency
- Make better use of underlying parallel hardware
- Take advantage of non-local resources

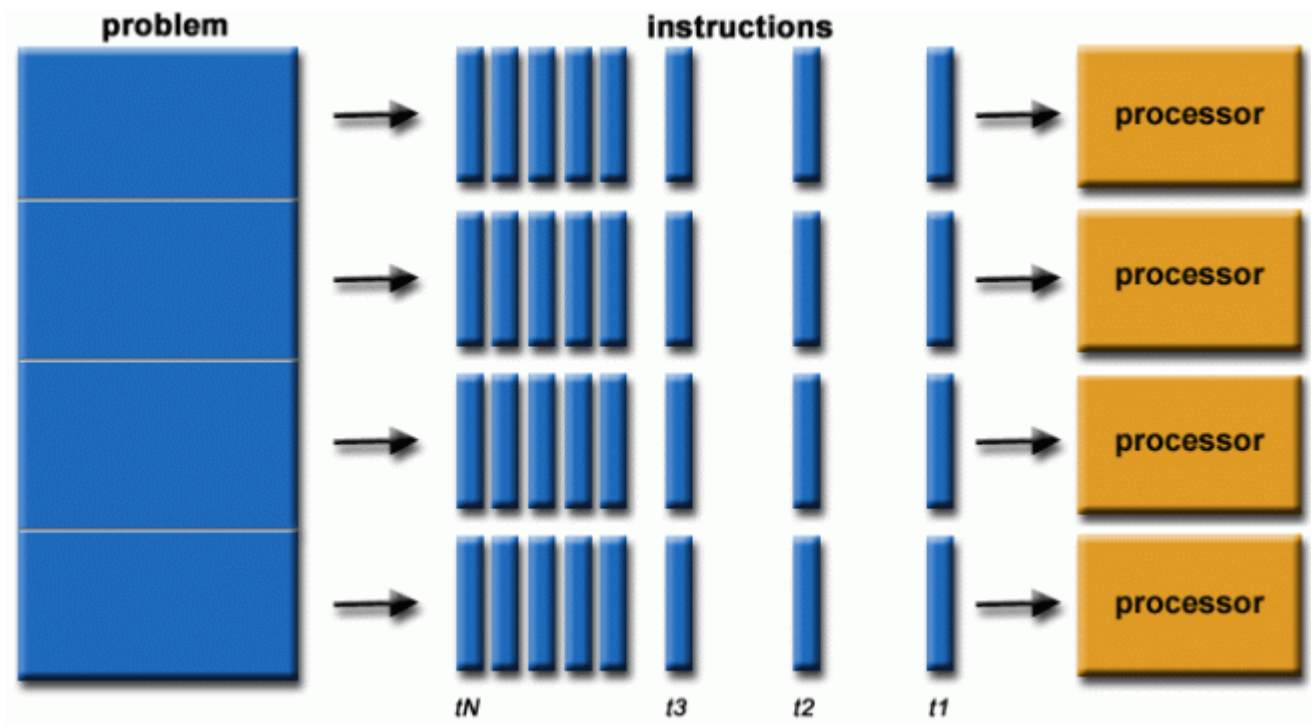
Parallelism is the necessity !!



Serial Computing



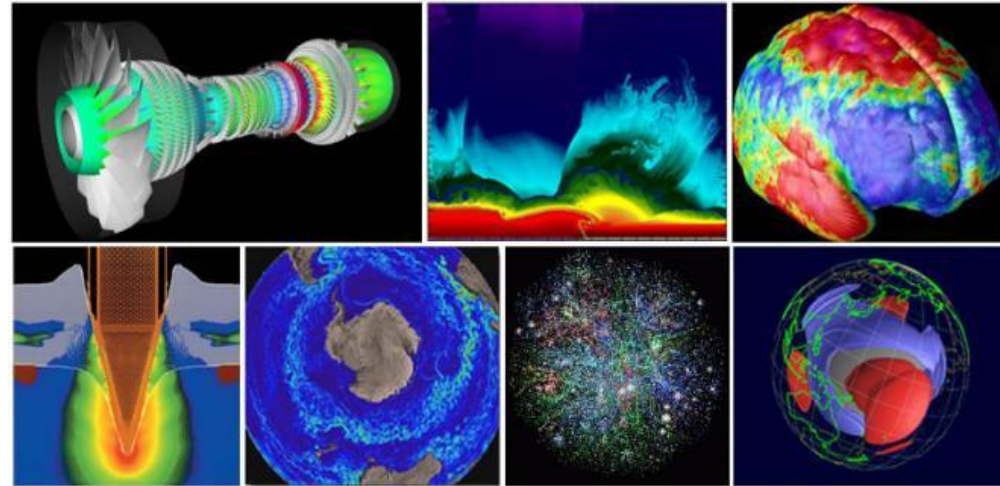
Parallel Computing



Uses of Parallel Computing

Science and Engineering

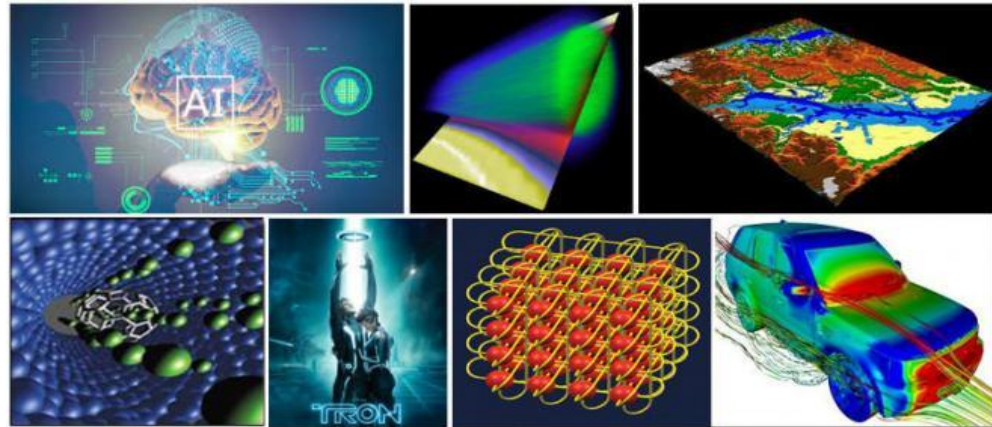
- Atmosphere, Earth, Environment
- Physics - applied, nuclear, particle, condensed matter, high pressure, fusion, photonics
- Bioscience, Biotechnology, Genetics
- Chemistry, Molecular Sciences
- Geology, Seismology
- Mechanical Engineering - from prosthetics to spacecraft
- Electrical Engineering, Circuit Design, Microelectronics
- Computer Science, Mathematics
- Defense, Weapons



Uses of Parallel Computing

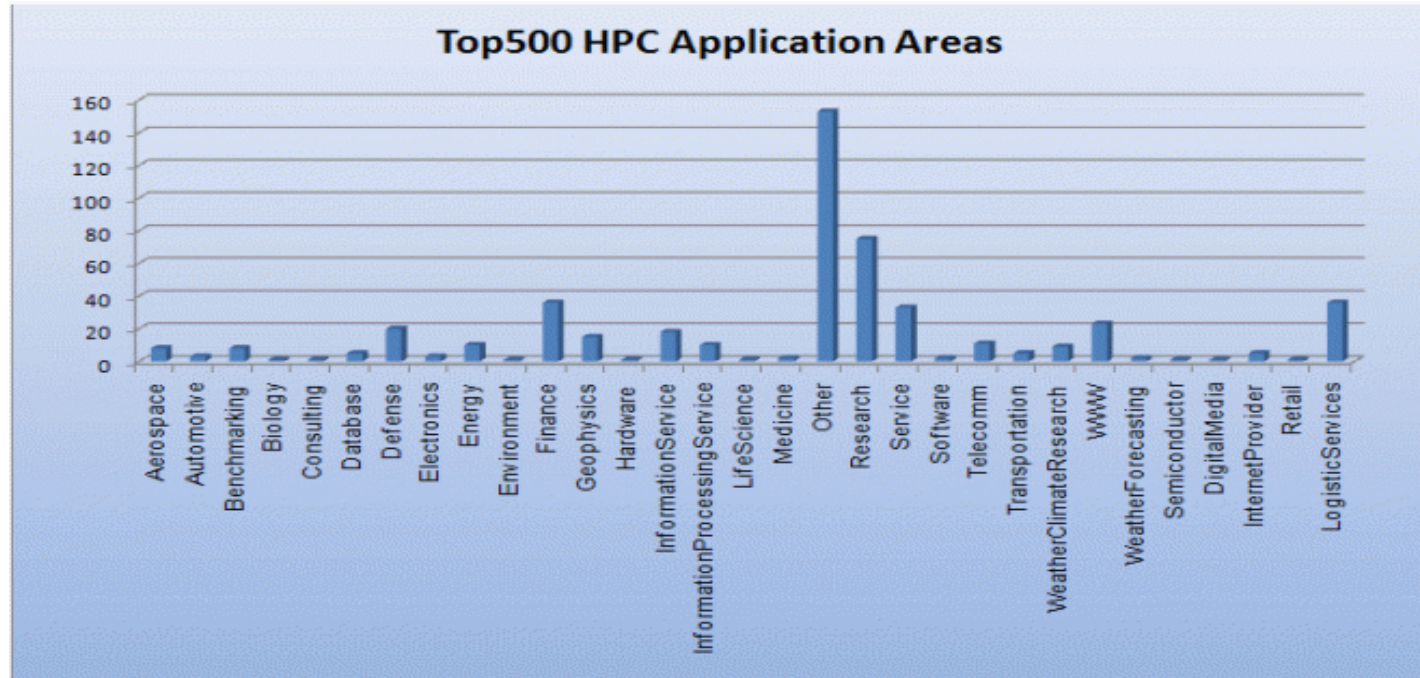
Industrial and Commercial

- Big Data, databases, data mining
- Artificial Intelligence (AI)
- Oil exploration
- Web search engines, web based business services
- Medical imaging and diagnosis
- Pharmaceutical design
- Financial and economic modeling
- Management of national and multi-national corporations
- Advanced graphics and virtual reality
- Networked video and multi-media technologies
- Collaborative work environments

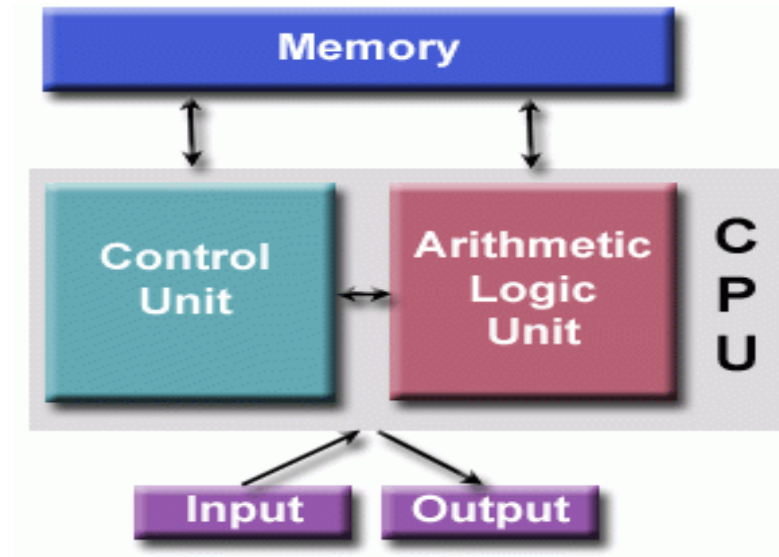


Uses of Parallel Computing

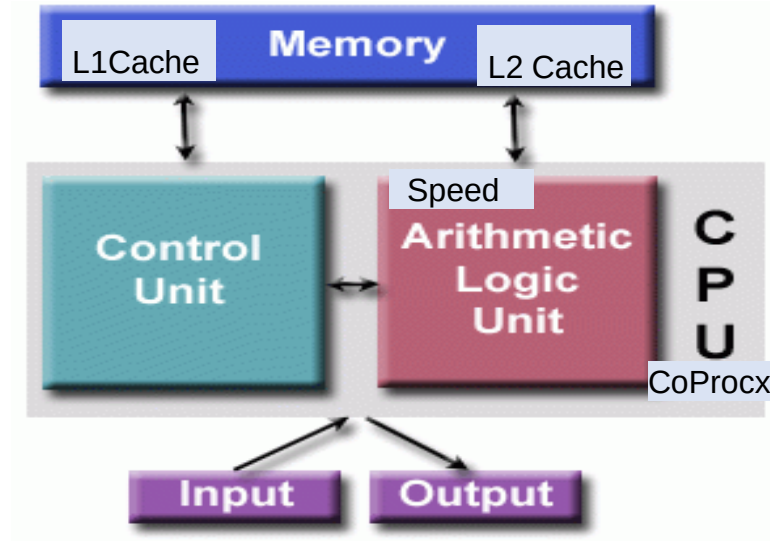
Global Applications



Simple Von Neumann Machine



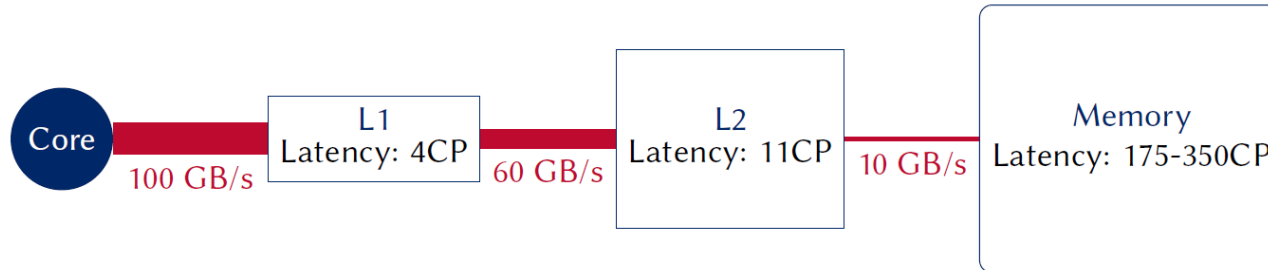
Von Neumann Architecture



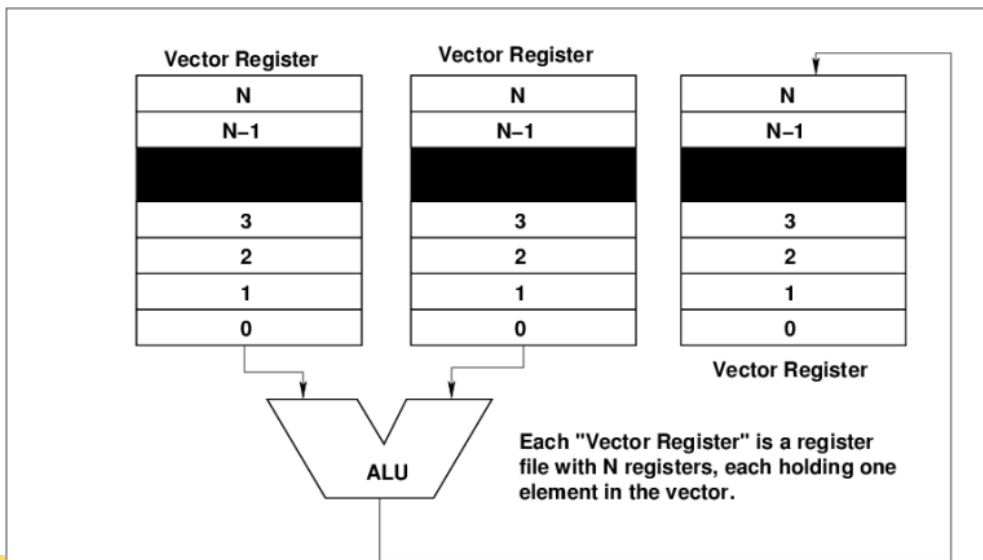
- Memory Hierarchy
 - Caches are smaller but faster (higher bandwidth, less latency) than memory
 - Multiple level of cache: L1, L2, L3
 - Helps keep cores fed with data
 - Cache reuse is critical to performance

Level	Size	Latency	Physical Location
L1 cache	32 KB	4 cycles	inside each core
L2 cache	256 KB	12 cycles	beside each core
L3 cache	6 MB	~21 cycles	shared between all cores
L4 E-cache	128 MB	~58 cycles	separate eDRAM chip
RAM	4+ GB	~117 cycles	SDRAM DIMMs on motherboard
Swap	100+ GB	10,000+ cycles	hard disk or SSD

Table 4 – The memory hierarchy of a modern desktop/laptop: Core i*4 Haswell.



- Vector Processors
 - able to run mathematical operations on multiple data elements simultaneously
 - Provides instruction that operate on entire array
 - Cray Machines



- Superscalar Processors
 - Instruction level parallelism with processor
 - Intel 960, SunSparc, MIPS R
 - and AMD Am29000

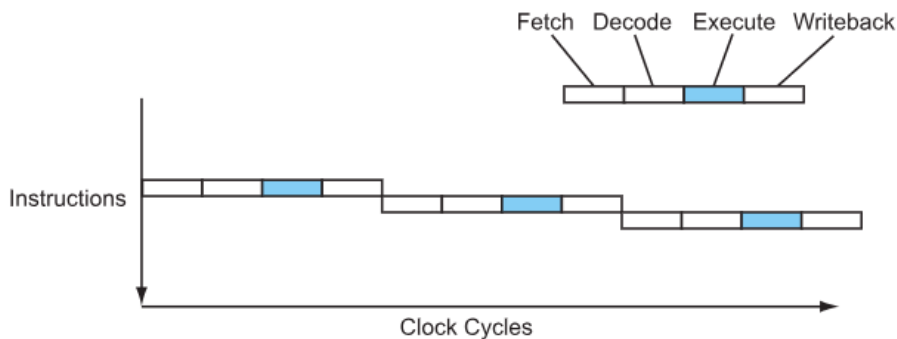


Figure 1 – The instruction flow of a sequential processor.

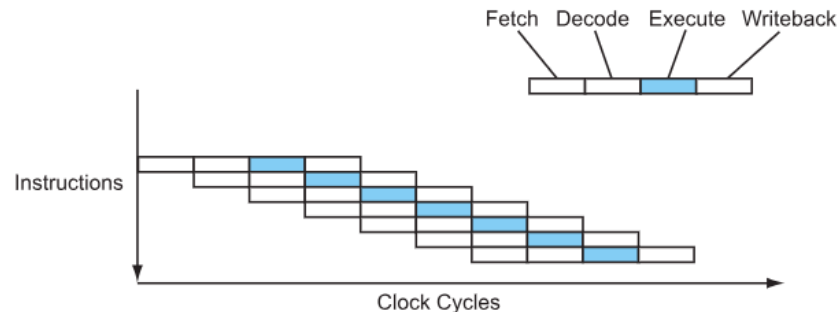
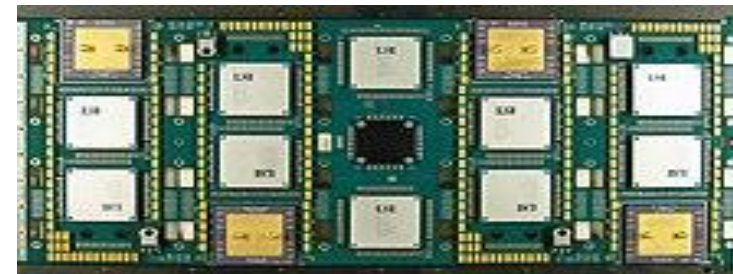


Figure 2 – The instruction flow of a pipelined processor.

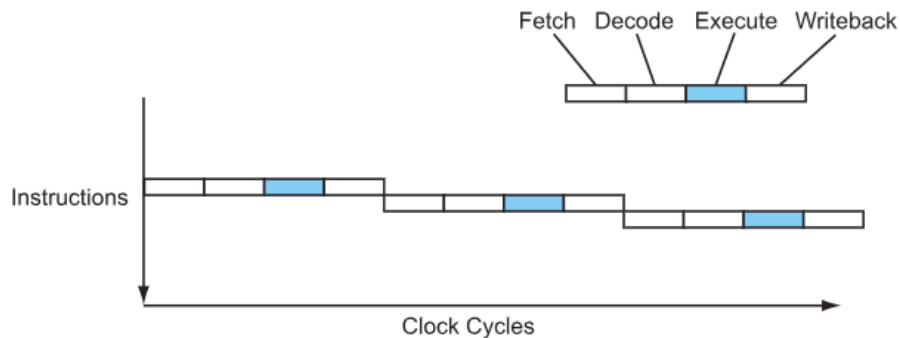


Figure 1 – The instruction flow of a sequential processor.

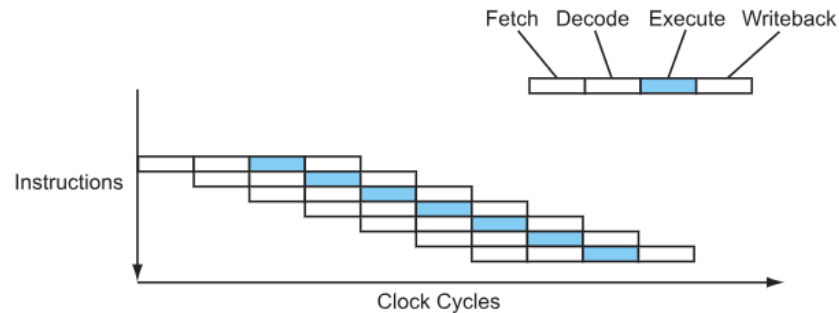


Figure 2 – The instruction flow of a pipelined processor.

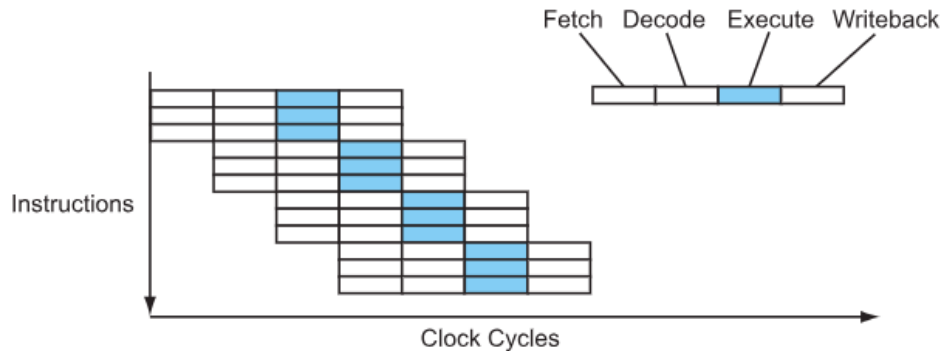
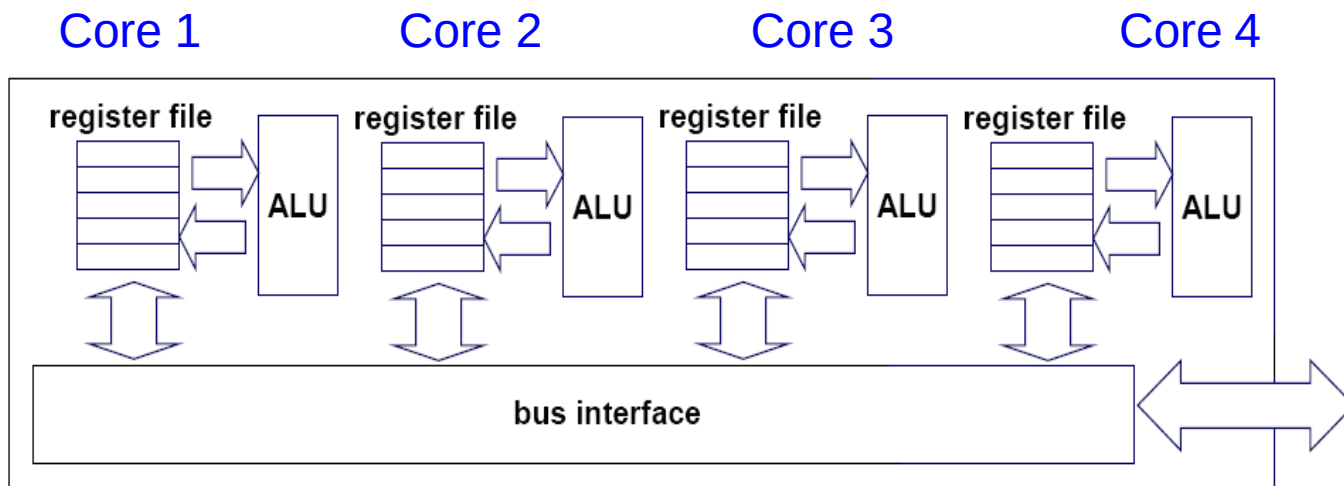
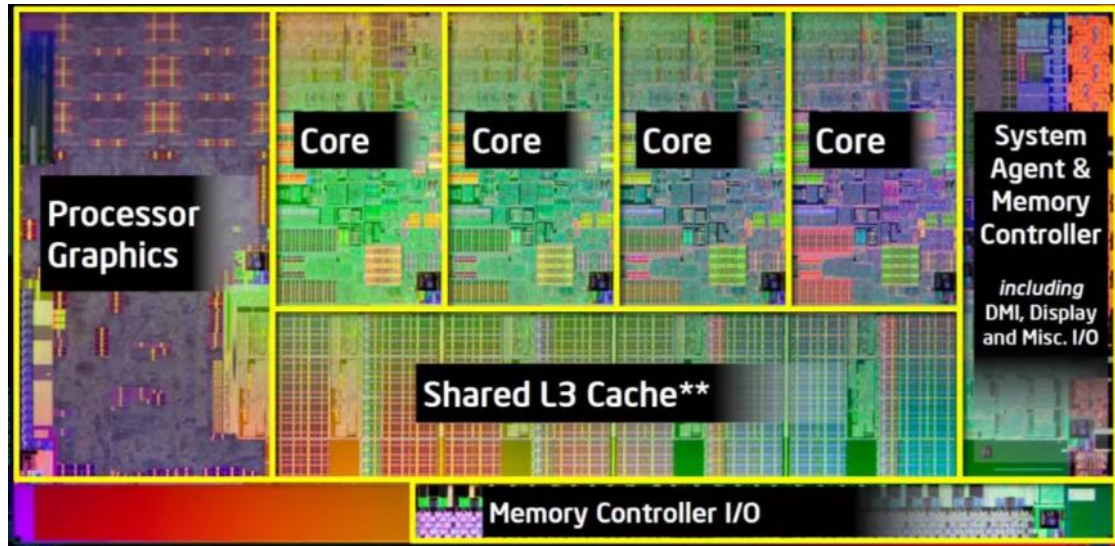


Figure 8 – The instruction flow of a superscalar processor.

- Core is a unit that reads and executes program instructions
- Processors were originally developed with only one core
- A multi-core processor is a single computing component with two or more independent processors ("cores").
- Replicate multiple processor cores on a single die.
- Manufacturers typically integrate cores onto a single integrated circuit die (chip multiprocessor or CMP), or onto multiple dies in a single chip package.







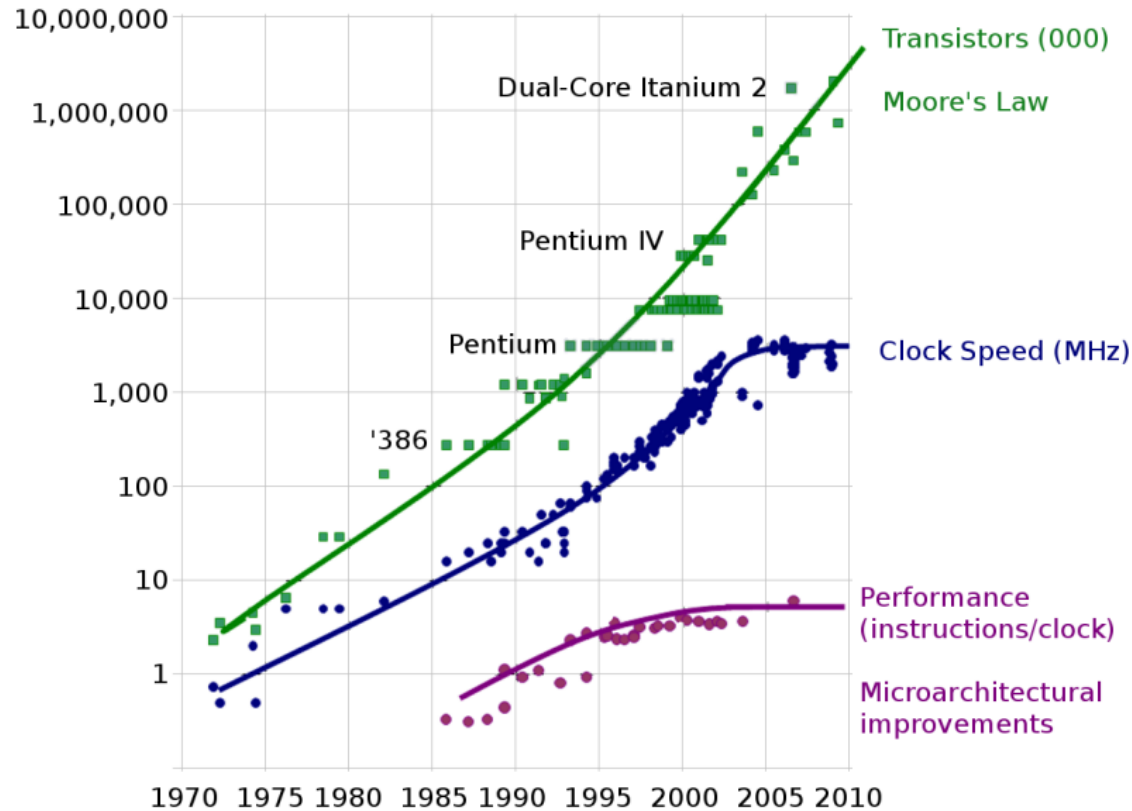
Why Multicores?



During 50 yrs of Moore's Law: transistor density X 2 every 2 years

- Transistors grew smaller
- Smaller transistors require less voltage and current
 - Can switch at higher frequency at same power
 - Clock rates increased
 - Capabilities could be added/expanded at constant power
 - Instruction-level parallelism
 - OO executions
 - Prefetching
 - More cache
- Single processor performance X 2 every 18 months

Intel CPU Trends



Sutter, *The Free Lunch is Over*, DDJ 2005.

Data: Intel, Wikipedia, K. Olukotun



Why Multicores?



Limits to single processor performance

- Moore's Law- CPU doubles its frequency every 2 yrs.
 - Possible by increasing the transistor density
 - Difficult to sustain because of heat produced
- Power increase with transistor size
- Power per transistor cannot be reduced
 - More transistors packed in same area -> higher power density -> more heat
 - Faster processors get too hot(~171F at 3.8GHz), cooling gets bulky and expensive
 - Speed has not really increased (~ 4 GHz max)
 - Adding lower frequency cores keeps computational capacity increasing at comparable power



Why Multicores?

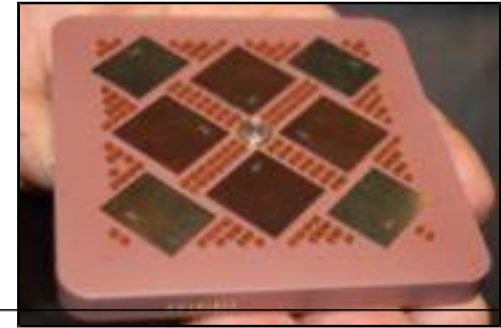


Logical extension is Multicore

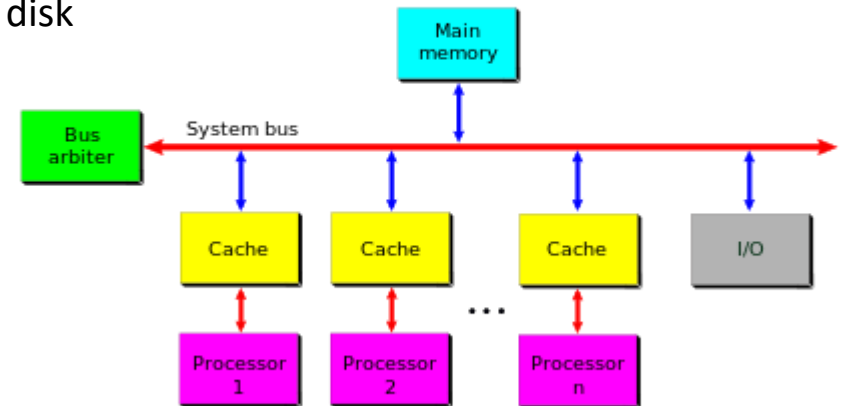
- 100s relatively slow(1.5GHz) processors vs 10s relatively fast(2.5GHz) processors
- Overall more compute available for comparable power
- Multicore is more general purpose
- A multi-core processor is an integrated circuit to which two or more processors have been attached for enhanced performance, reduced power consumption and more efficient simultaneous processing of multiple tasks.

Symmetric Multiprocessors (SMP)

- Two or more identical/ homogenous processors are connected to a single shared main memory
- Inter-processors communication takes place through shared memory
- Shared memory hardware architecture where multiple processors share a single address space and have equal access to all resources - memory, disk

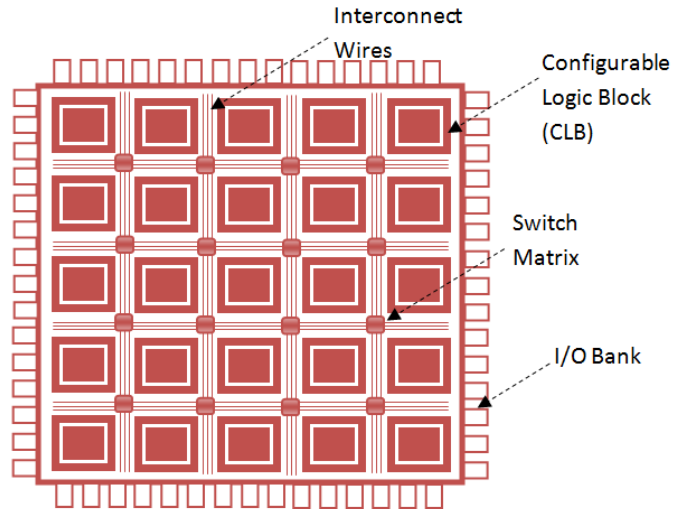


SMP - symmetric multiprocessor system



Field-Programmable Gate Arrays

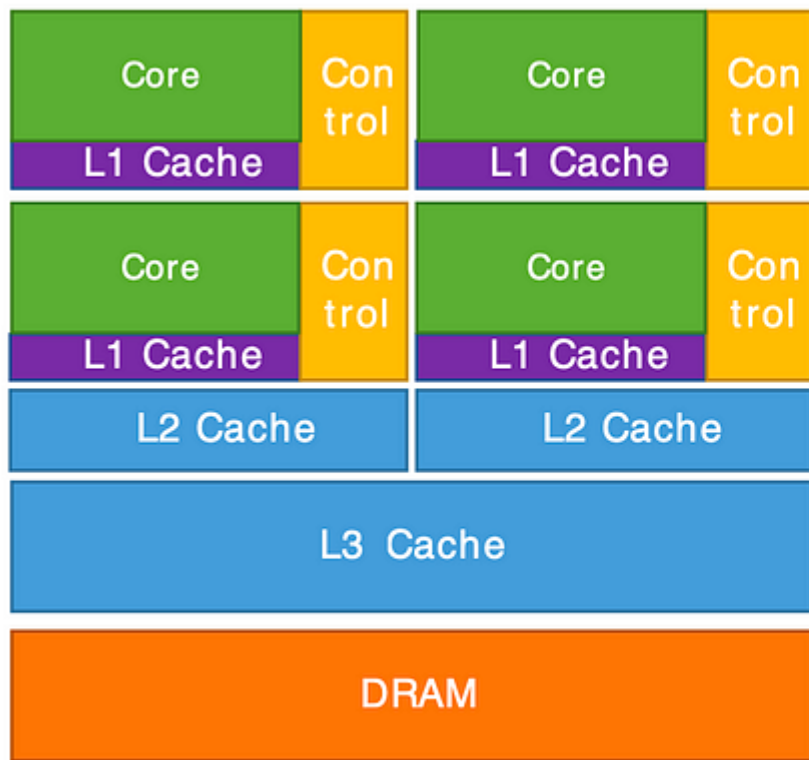
- a computer chip that can rewire itself for a given task
- can be programmed with hardware description languages such as VHDL or Verilog



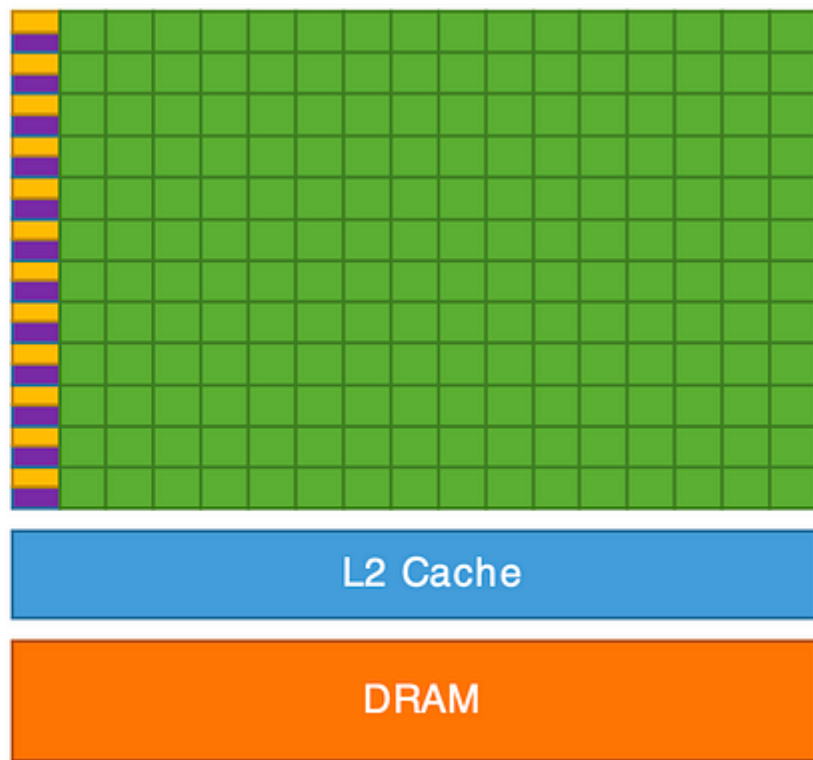
General Purpose GPU

- General-purpose computing on graphics processing units (GPGPU)
- NVIDIA, AMD and Intel
- Nvidia GPU cores are known as streaming multiprocessors.
- CUDA/OpenCL programming environment





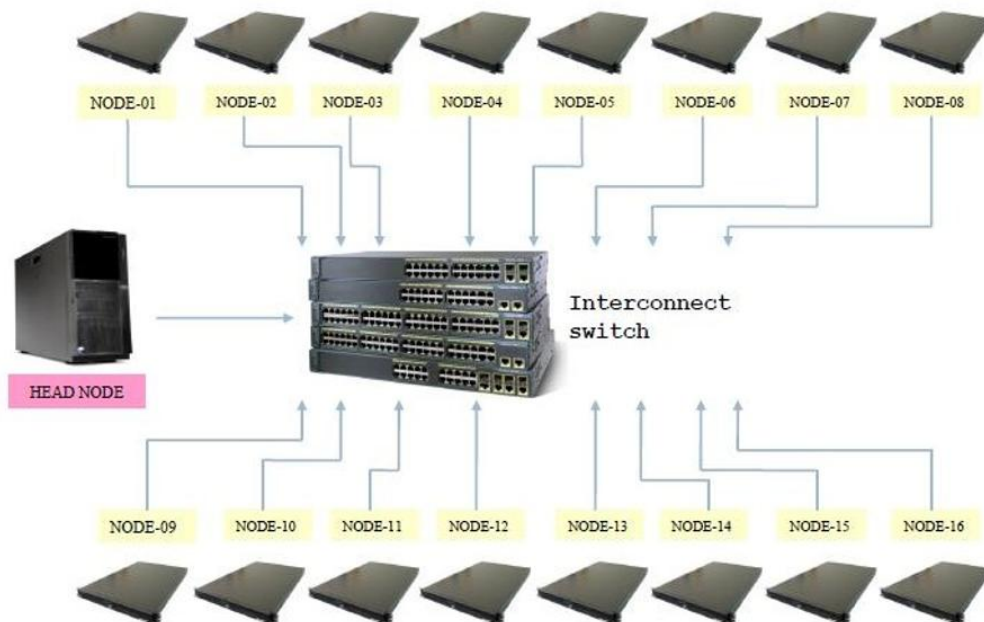
CPU



GPU

- Tightly coupled computers that work together closely as though they are a single computer.

Clusters



Supercomputer Clusters

- Tightly coupled computers that work together closely as though they are a single computer.





Cluster Computing



Processing

- In general, there are two parallel processing approaches: symmetric multiprocessing (SMP) and massively parallel processing (MPP).

Interconnects

- In order for a large number of processors to work together, supercomputers utilize specialized network interfaces. These interconnects support high bandwidth and very low latency communication.
- Interconnects join nodes inside the supercomputer together.
- Popular Interconnects include GigE & Infiniband

Operating Systems

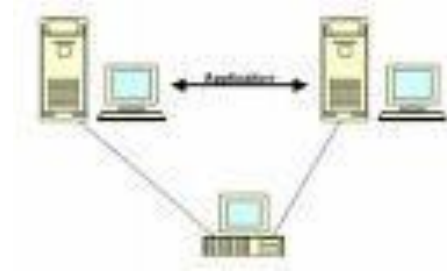
- Supercomputers today most often use variants of the Linux operating system.
- "lightweight" operating systems that consist of a small, simple kernel with many of the capabilities of a general-purpose O/S removed.

Programming

- The parallel architectures of supercomputers demand use of special programming techniques to exploit their speed.
- The base language of supercomputer code is, in general, Fortran or C, using special libraries to share data between nodes.
- Environments such as MPI and PVM for loosely connected clusters and OpenMP for tightly coordinated shared memory machines are used.
- GPGPUs have hundreds of processor cores and are programmed using programming models such as CUDA and OpenCL.

Distributed Computing

- different parts of a program are run simultaneously on separate computers communicating over a network



Grid Computing

- Aggregation, sharing of distributed heterogeneous systems



Cloud computing

- on-demand available service- IaaS, PaaS, SaaS
- pay-as-you-use over the internet

